View and Viewpoints

Module 12

Module Objectives

- To understand the concepts of views and viewpoints
- To understand the role of Architecture Views
- To introduce some TOGAF resource



Concepts and Definitions

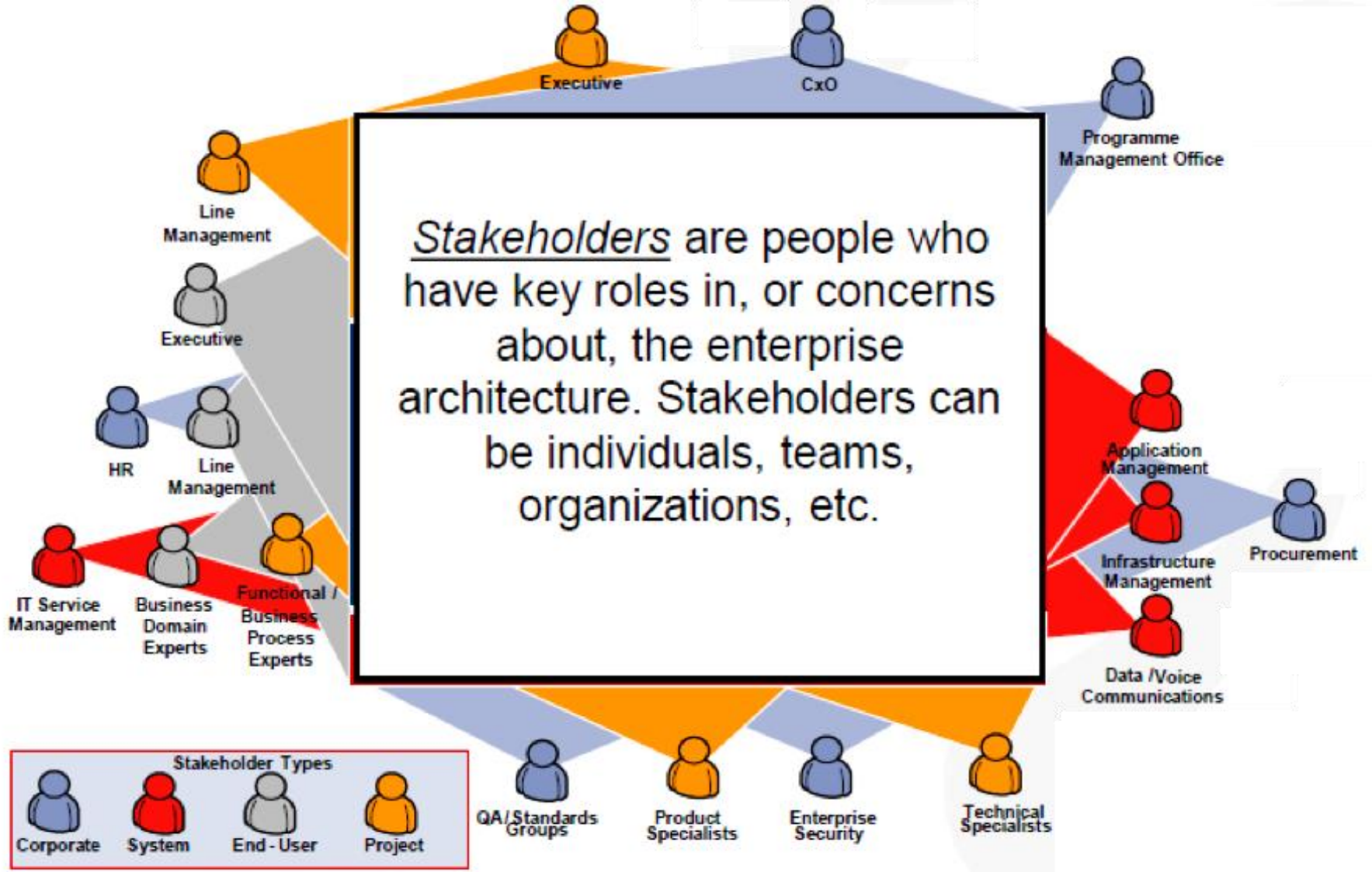
- System
- Stakeholders
- Concerns
- View
- Viewpoint

System

- **A system** is a **collection of components** organized to accomplish a specific function or a set of **functions**

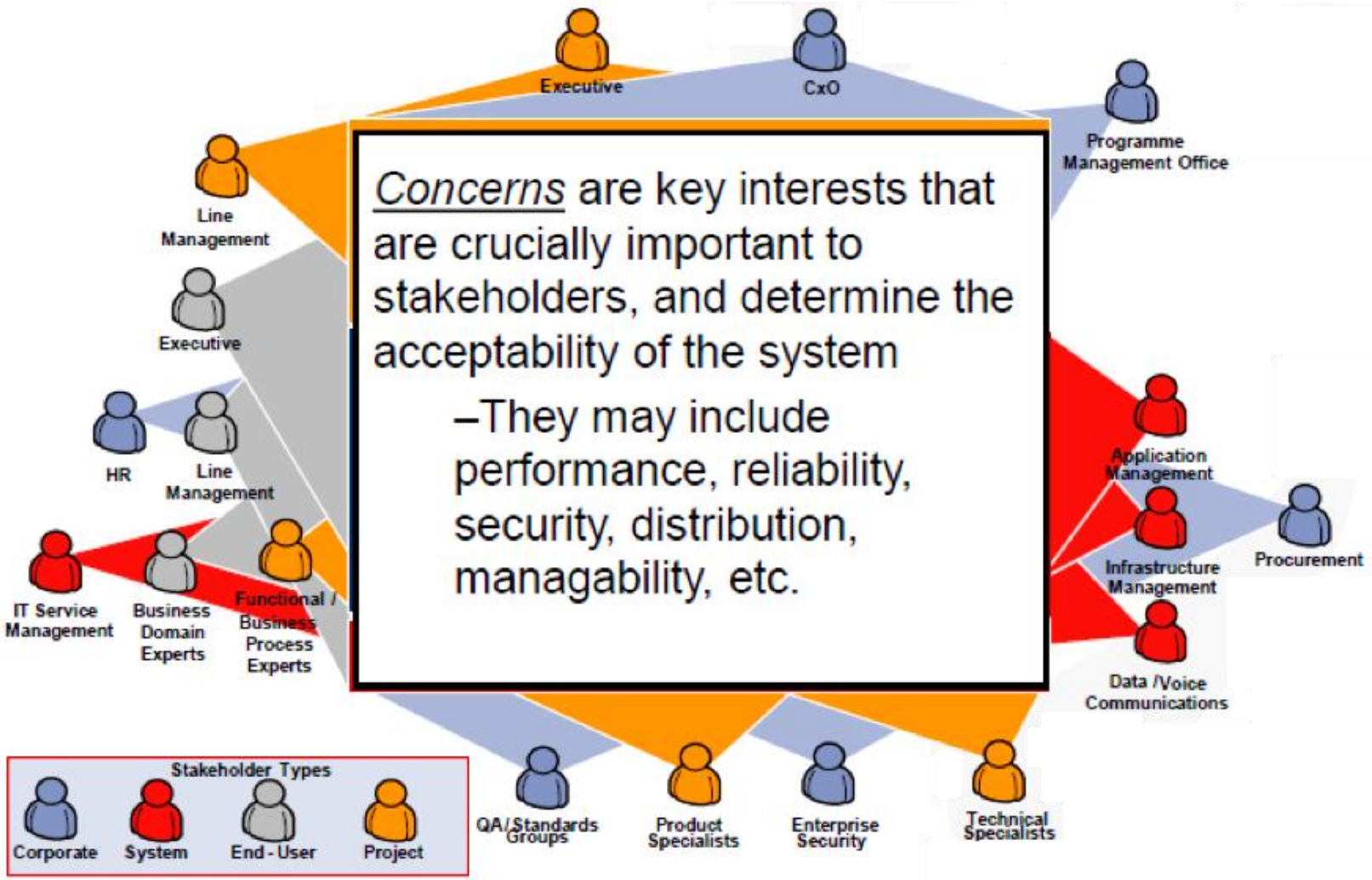


Stakeholders



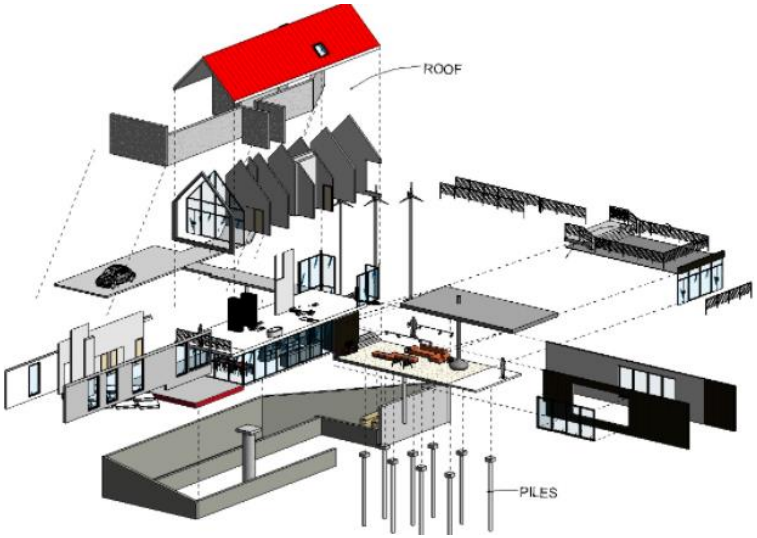
Stakeholders are people who have key roles in, or concerns about, the enterprise architecture. Stakeholders can be individuals, teams, organizations, etc.

Concerns


Executive
CxO
Programme Management Office
Line Management
Executive
HR
Line Management
IT Service Management
Business Domain Experts
Functional / Business Process Experts
Application Management
Infrastructure Management
Procurement
Data /Voice Communications
QA/Standards Groups
Product Specialists
Enterprise Security
Technical Specialists
Concerns are key interests that are crucially important to stakeholders, and determine the acceptability of the system
–They may include performance, reliability, security, distribution, managability, etc.
Stakeholder Types
Corporate
System
End-User
Project

View

- **A view** is a **representation** of a system from the **perspective** of a related **set of concerns**
  - An architect creates architecture models. A view consists of parts of these, chosen to show stakeholders that their concerns are being met

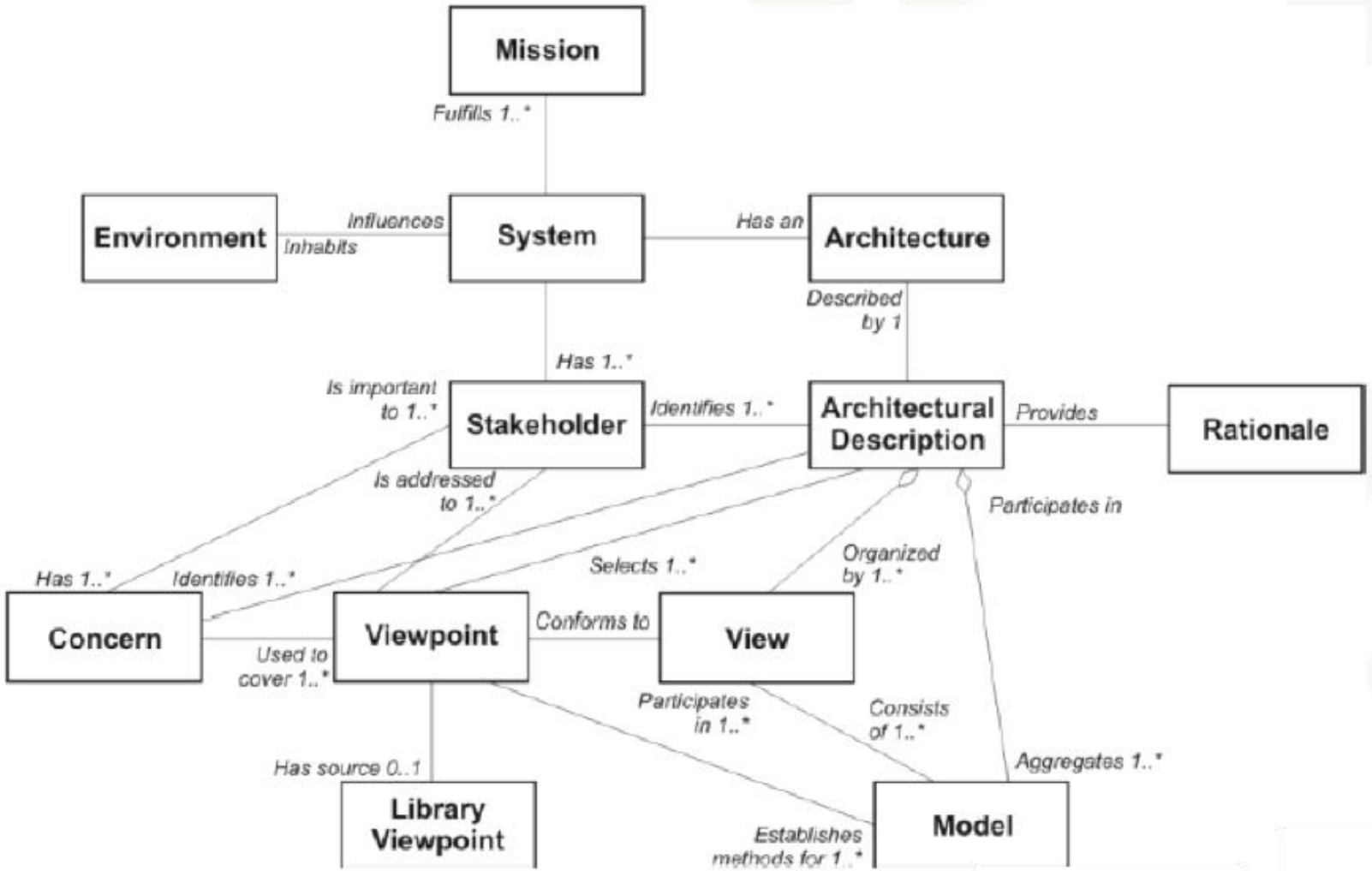


Viewpoint

- **A viewpoint** defines the **perspective** from a which a **view is taken**
  - It defines **how to construct and use a view**, the information needed, the modeling techniques for expressing and analyzing it and a rationale for these choices (e.g. by describing the purpose and intended audience of the view)



ISO/IEC 42010:2007



Source: IEEE Std 1471-2000. Used with permission

View and Viewpoints

The architect uses views and viewpoints in **Phases A to D** for developing architectures for each domain

- A view is **what you see**
- A viewpoint is **where you are looking from**, the vantage point or perspective that determines what you see
- Every view has an associated viewpoint that describes it, at least implicitly
- Viewpoints are generic, and can be stored in libraries for reuse. A view is **always specific** to the architecture for which it is created

What is an Architecture View?

- **A representation of an overall architecture** with meaning to one or more stakeholders in the system
- E.g. a building architect might create wiring diagrams, floor plans, and elevations to describe different faces of a building to its different stakeholders
- An enterprise architect might create **physical** and **security views** of an IT system

A Simple Example of a Viewpoint

Viewpoint Element	Description
Stakeholders	Management
Concerns	Show the top-level relationships between geographical sites and business functions
Modeling Technique	Nested boxes diagram
	Outer boxes = locations
	Inner boxes = business functions
	Semantics of nesting = functions performed in the locationl

A Simple Example of a View

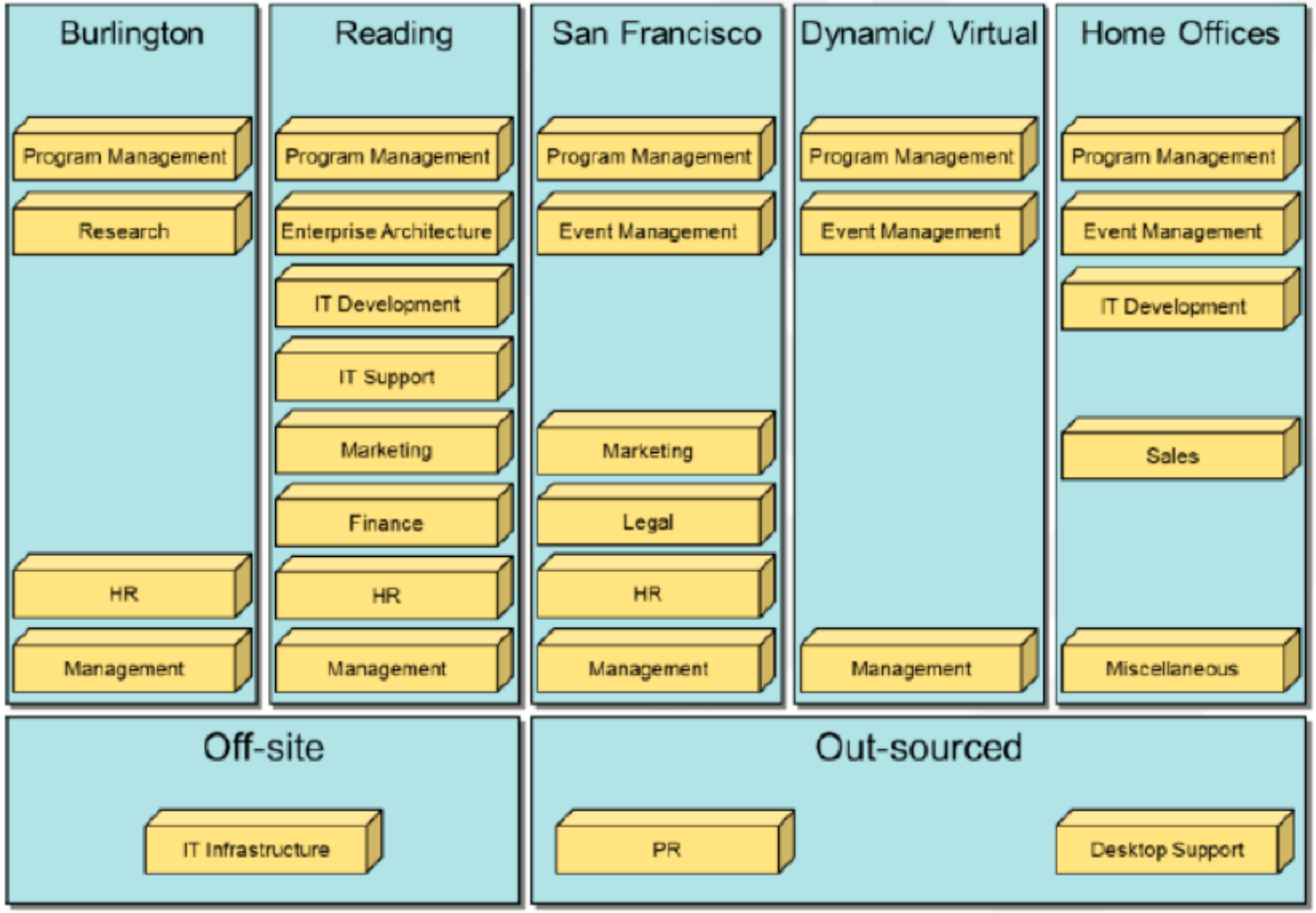


Figure 1: Example View - The Open Group Business Domains in 2008

Developing Views in the ADM

- The choice of which particular **architecture views** to develop is one of the key decisions that the architect has to make
- The architect has a responsibility for ensuring
  - the **completeness** of the architecture
  - does it address all the **concerns of its stakeholders**
- The integrity of the architecture
  - can the views be connected to each other?
  - can the conflicting concerns be reconciled?
  - what trade-offs have been made?

The View Creation Process

1. Refer to any **existing libraries** of viewpoints
2. Select key **stakeholders**
3. Analyze their **concerns** and document them
4. Select appropriate **viewpoints** (based on the stakeholders and their concerns)
   - If no libraries of viewpoints exist, develop **new** viewpoints
5. Generate **views** of the system using the selected viewpoints as a template

Benefits

- **Less work** for architects (the viewpoints have already been defined and so the views can be created faster)
- **Better comprehensibility** for stakeholders (the viewpoints are already familiar)
- **Greater confidence** in the validity of the views (their viewpoints have a known track record)

Using TOGAF Artifacts

- TOGAF includes an example set of recommended **artifacts** that can be adopted, enhanced and combined to produce **architecture views**
- Three classes of artifacts are defined
  - Catalogs
  - Matrices
  - Diagrams

Catalogs

- Catalogs are **lists of building blocks** of a specific type, or of related types
- For example
  - Principle Catalog in Preliminary Phase
  - Organization/Actor Catalog in Phase B
  - Driver/Goal/Objective Catalog

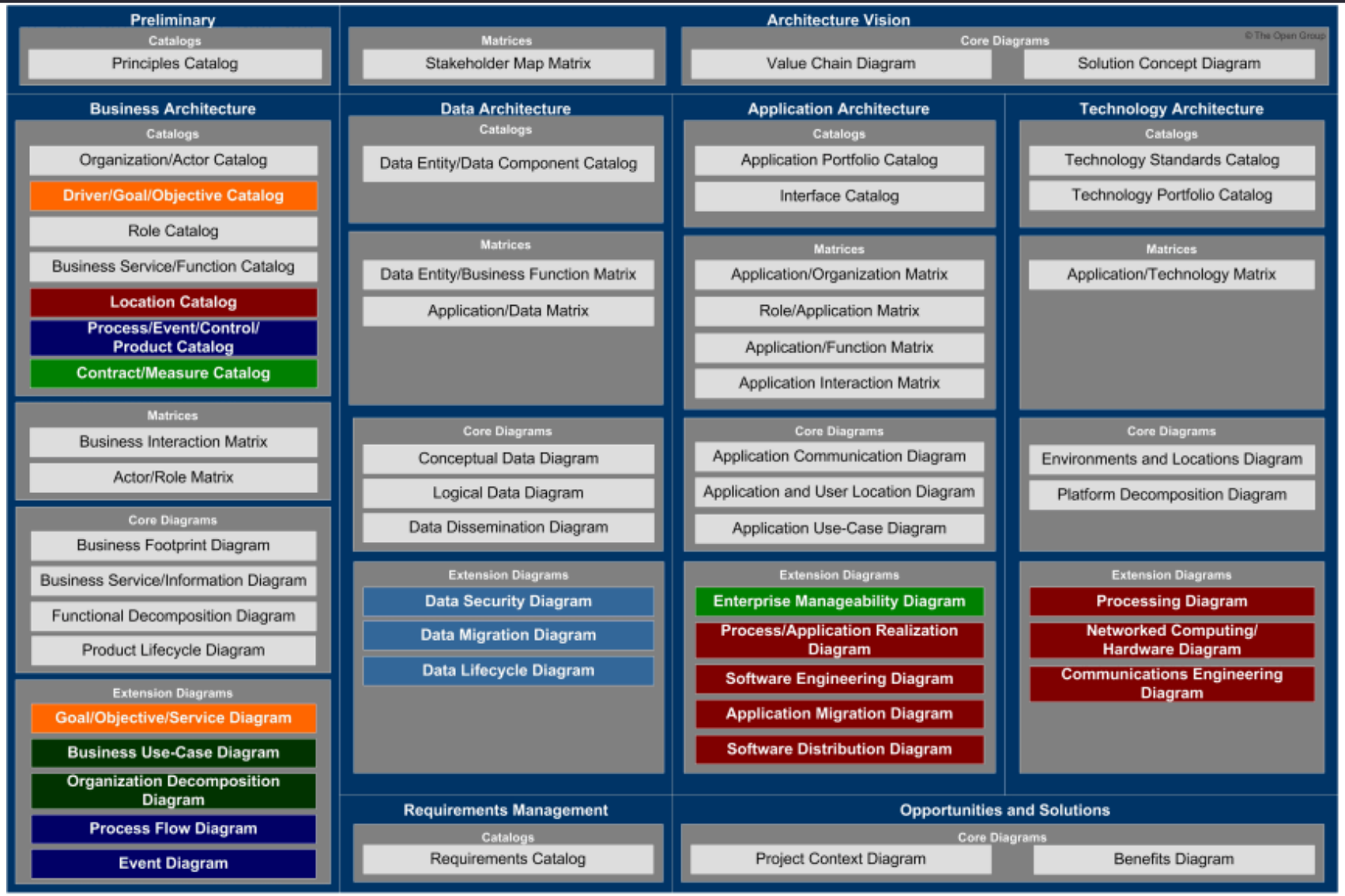
Matrices

- Matrices show **the relationship** between building blocks of specific types
- Matrices are used to represent **list-based** rather than graphical-based relationships
- For example
  - The Stakeholder Map Matrix in Phase A

Diagrams

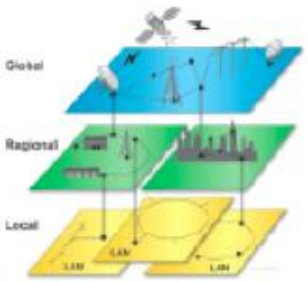
- Diagrams representing building blocks in a rich and **visual way,** especially suited to **stakeholder communication**
- For example
  - Value Chain Diagram in Phase A
  - Business Footprint Diagram in Phase B

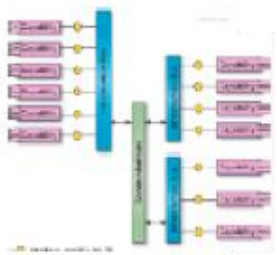
TOGAF 9 Artifacts

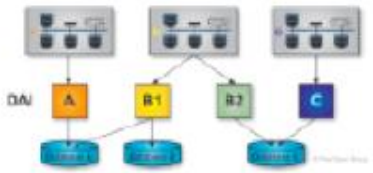


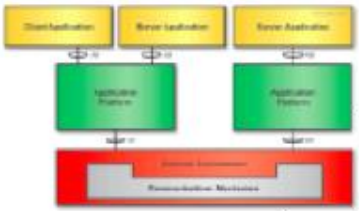
THE Open GROUP

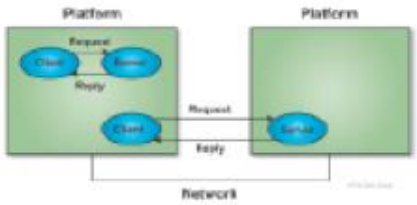
Recommended Architecture Views













- Business Architecture View
- Enterprise Security View
- Software Engineering View
- Communications Engineering View
- Data Flow View
- Enterprise Manageability View
- Acquirer View

Summary

- In general, TOGAF embraces the concepts and definitions of **ISO/IEC 42010:2007**, specifically those guiding the development of a view and make it actionable, such as
  - Selecting **key stakeholders**
  - Analyzing their **concerns** and documenting them
  - Understanding **how to model** and deal with those concerns
  - The language used to depict the view is the viewpoint. Viewpoints provide **architecture concepts** from different perspectives, including components, interfaces, and allocation of services critical to the view

Summary

When applying TOGAF, a number of tailoring steps should occur:

- The **viewpoints** provided should be **customized** to create a set of **architecture views** that ensure all stakeholder concerns are met
- **New viewpoints** and **views** should be created to address specific needs